

PHASE 5 OF 8

Frontend, Preview & Editing

Milestones 4 and 5: the Streamlit interface, how input reaches the backend, and how the draft is reviewed and edited.

Milestone 4

Milestone 5

Streamlit

Session State

I Milestone 4 — Frontend Development

Activity 4.1: Streamlit Configuration

frontend/app.py — page configuration:

```
import os
import requests
import streamlit as st
from dotenv import load_dotenv

load_dotenv()

BACKEND_URL = os.getenv(
    "BACKEND_URL",
    "http://127.0.0.1:8000"
).rstrip("/")

st.set_page_config(
    page_title="LegalEase",
    page_icon="⚖️",
    layout="wide"
)
```

The backend address comes from BACKEND_URL (defaulting to the local FastAPI server), and `rstrip("/")` prevents a double slash when the route is appended. The wide layout gives the preview and editor room to sit side by side.

Activity 4.2: Sending User Input to the Backend

```
payload = {
    "document_type": document_type,
    "parties": parties,
    "effective_date": str(effective_date),
    "jurisdiction": jurisdiction,
    "terms": terms,
    "additional_instructions": additional_instructions,
}
response = requests.post(f"{BACKEND_URL}/generate", json=payload, timeout=120)
response.raise_for_status()
document = response.json()["document"]
```

The frontend collects the form values into a JSON payload, sends them to FastAPI and displays the returned document.

The main interface

On first load the page shows the LegalEase title, the tagline, an amber notice that outputs are AI-generated drafts, and two empty panels waiting for a document.

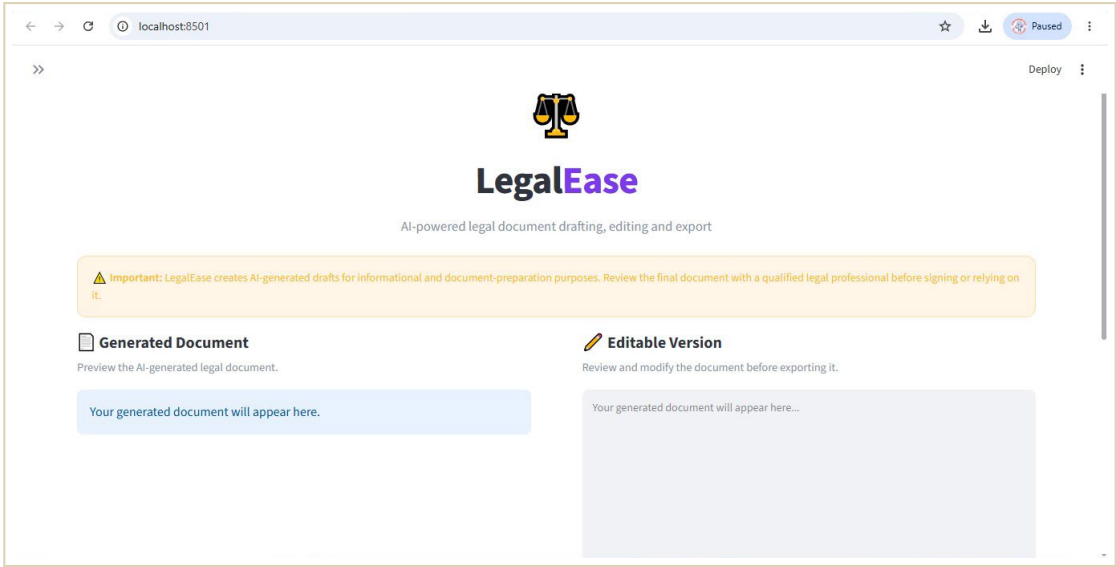


Figure 1 — LegalEase main user interface.

The Document Setup form

The sidebar collects everything the backend needs. The screenshot shows the demonstration values already filled in.

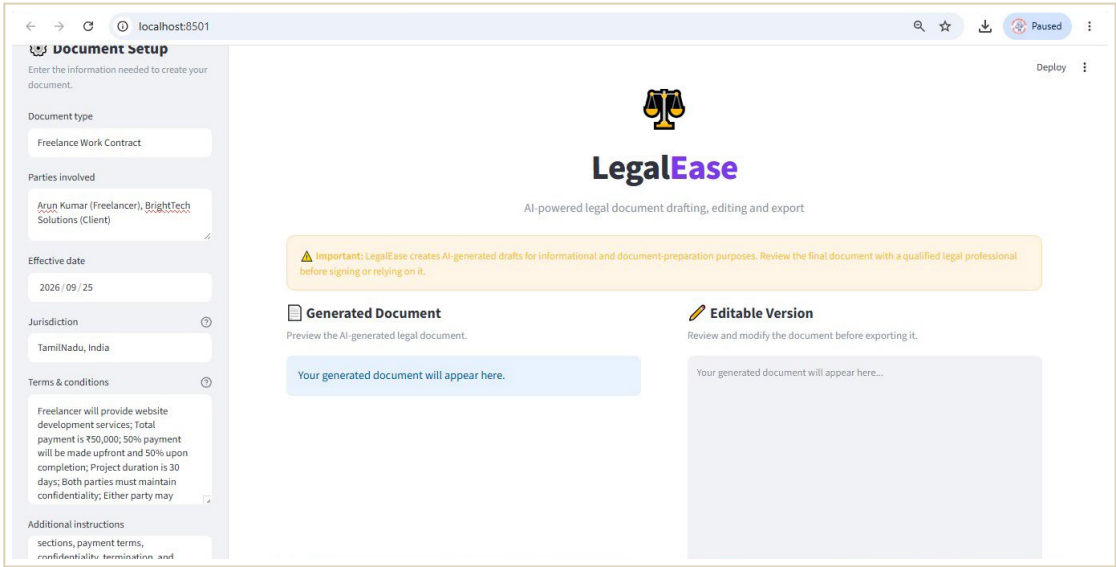


Figure 2 — Document Setup and input form.

Exploring the Input Fields

Field	Purpose	Demonstration value
Document type	Identifies the type of document to generate.	Freelance Work Contract
Parties involved	Identifies the people or organizations in the agreement.	Arun Kumar (Freelancer), BrightTech Solutions (Client)
Effective date	Specifies the agreement's effective date.	2026/09/25
Jurisdiction	Specifies the requested jurisdiction.	Tamil Nadu, India
Terms & conditions	Substantive requirements: services, payment, duration, confidentiality, termination.	Website development; ₹50,000; 30 days...
Additional instructions	Extra drafting or formatting requirements.	Section, payment-term and confidentiality guidance

Milestone 5 — Document Preview & Editing

Activity 5.1: Generated Preview

The generated text is stored in Streamlit **session state** and displayed in the *Generated Document* preview. A separate editable text area lets the user modify the draft before export.

Activity 5.2: Editable Version

```
edited_document = st.text_area(
    "Editable Version",
    value=st.session_state.document,
    height=500,
    key="edited_document",
)
```

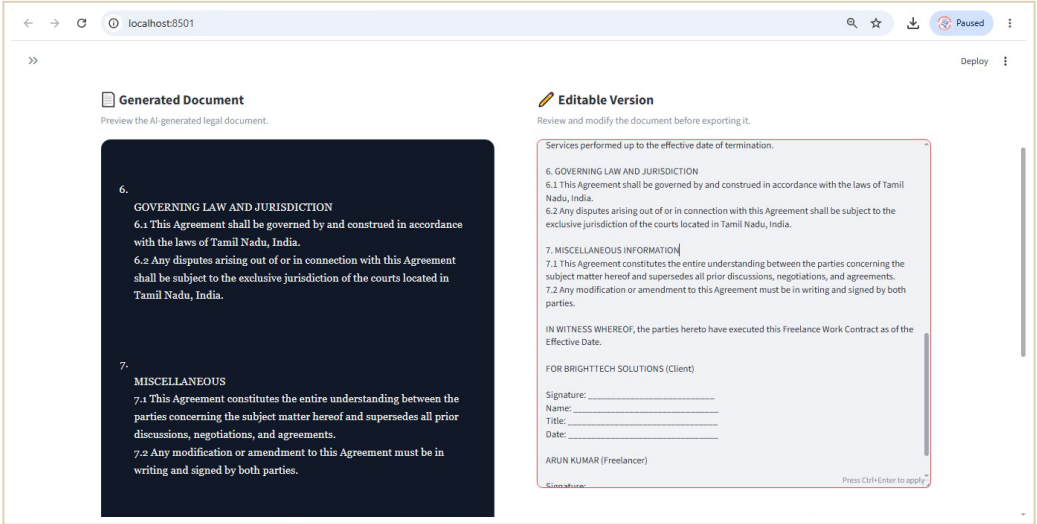


Figure 4 — Editable legal document preview.

Because the text area is keyed and seeded from session state, edits survive Streamlit's automatic reruns, and the exports use the *edited* text rather than the original AI output.